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INSTALLATION GUIDE

Debian 12, Apache, PostgreSQL,
and PHP

PRESENTED BY

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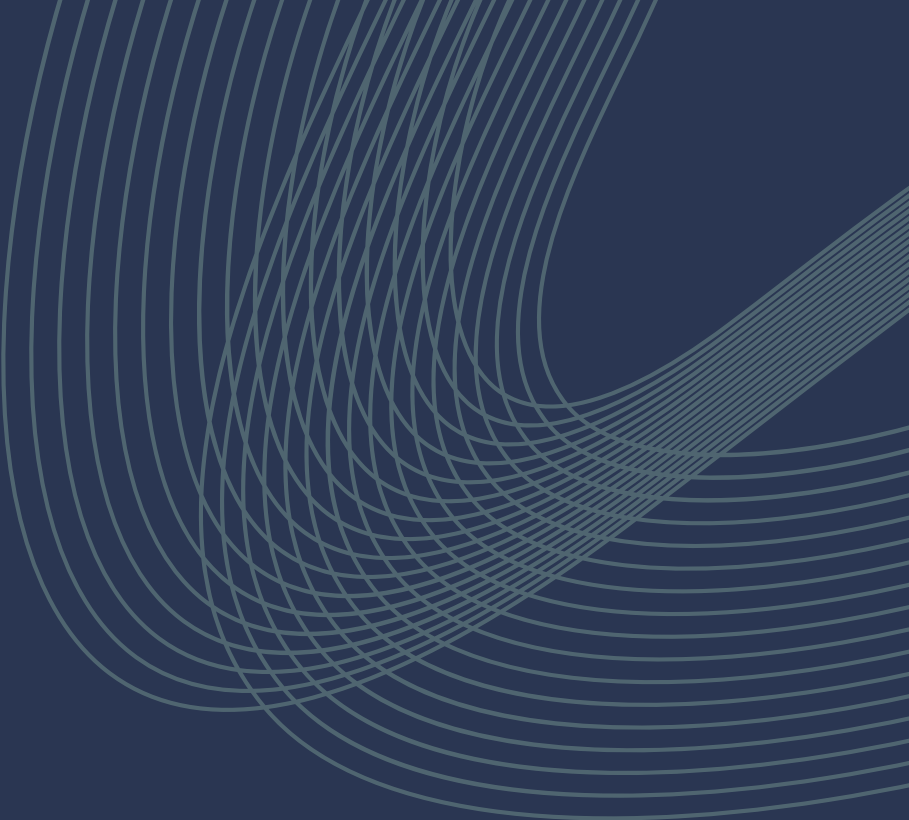


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INTRODUCTION

This guide walks you through the installation of Debian 12 (Bookworm) without a graphical interface, Apache web server, PostgreSQL database, and PHP on a virtual machine using QEMU/KVM. The installation is performed in a command-line environment, ensuring a lightweight and secure setup for web development purposes.

0. Prerequisites



A machine with QEMU/KVM installed.



Debian 12 ISO image (netinst) available in the directory `/usr/local/images-ISO/`.



You can download the Debian 12 ISO for `x86_64` architecture from the following link: [Debian ISO download](#)



Verify the Integrity of the ISO Image: To verify the integrity of the ISO image, use the `sha512sum` command to check the checksum:

```
SHA512SUM /USR/LOCAL/IMAGES-ISO/DEBIAN-12.X-NETINST.ISO
```



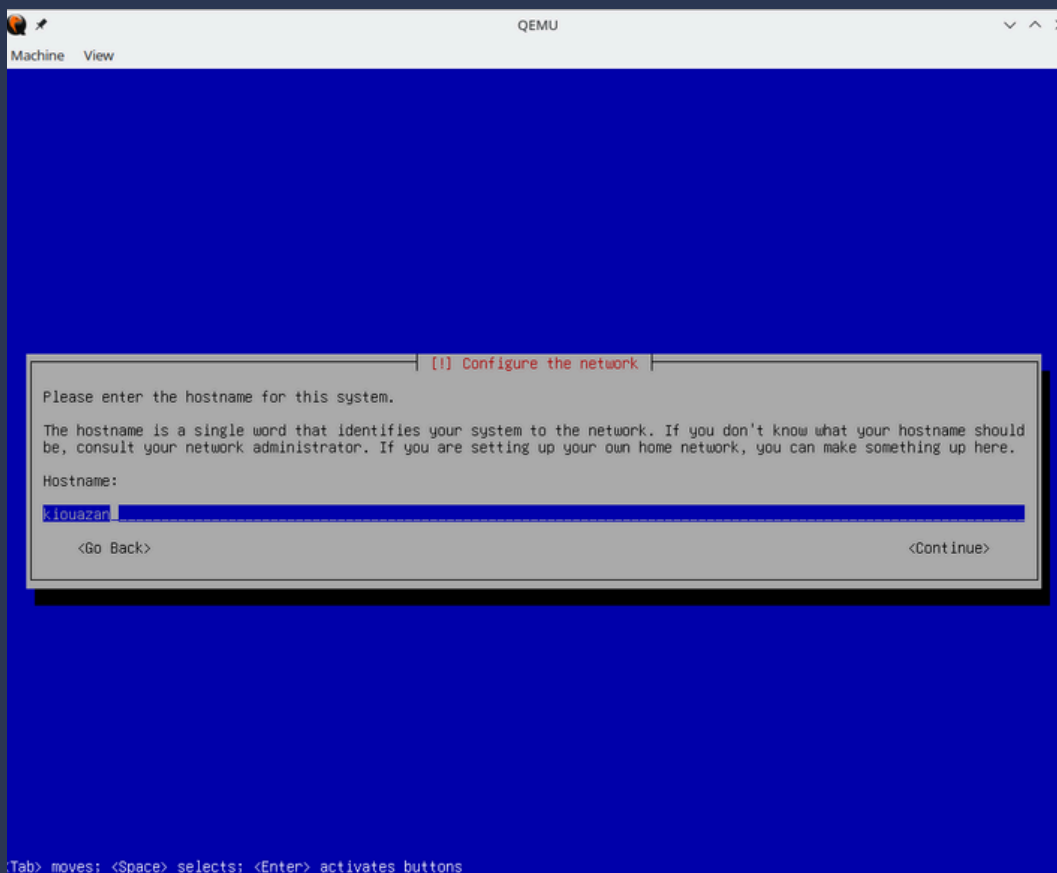
1.LET'S START

Launch the Virtual Machine

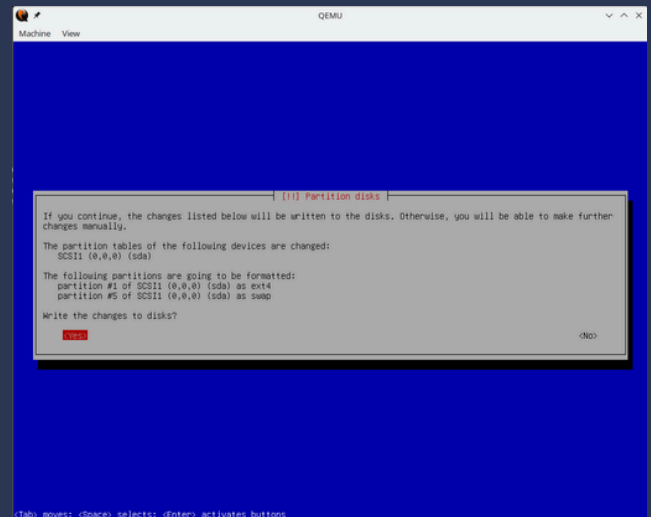
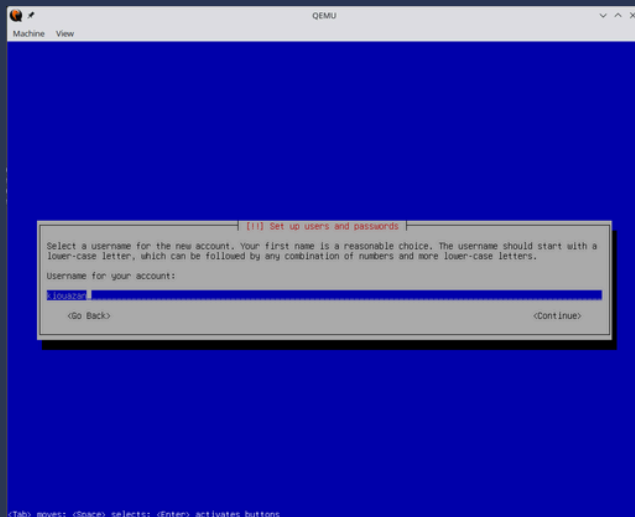
START THE VIRTUAL MACHINE WITH THE DEBIAN INSTALLATION ISO: USE THE FOLLOWING SCRIPT OR COMMAND TO LAUNCH QEMU/KVM:

Follow the Installation Process:

- Language: Select English.
- Location: Choose other/Europe/France.
- Locales: Set to United States, en_US.UTF-8.
- Keyboard: Select French.
- Hostname: Enter server-<YOUR_UGA_LOGIN>
(replace <YOUR_UGA_LOGIN> with your UGA login).



1.1 DISK PARTITIONING, SOFTWARE SELECTION & USER SETUP



- **Partition Disks:** Select Guided - use entire disk.
 - Choose All files in one partition.
 - Confirm with Yes.
- **Software Selection:** Ensure that Debian desktop is not selected and SSH server is selected.
- **Install GRUB:** Choose Yes.
- **Device for Boot Loader:** Select `/dev/sda`.
- **Root Password:** Choose a simple password like root.
- **User Account:**
 - **Full Name:** Enter your full name (e.g., Jean Toto).
 - **User Name:** Enter your UGA login name.
 - **User Password:** Set a simple password like etu.



2. TRANSFER AND PRESERVE THE DISK IMAGE

- ENSURE THE VIRTUAL MACHINE IS POWERED OFF.
- PRESERVE THE DISK IMAGE BY SAFELY TRANSFERRING IT TO THE TARGET SERVER.
- USE THE APPROPRIATE COMMAND TO MOVE THE DISK IMAGE TO THE SERVER (E.G., EREBUS4 IN THIS CASE).

VERIFICATION : CAPTURE THE OUTPUT OF THE FOLLOWING COMMAND INSIDE YOUR VIRTUAL MACHINE: `CAT /ETC/FSTAB`. THIS CONFIRMS THAT THE FILESYSTEM IS PROPERLY CONFIGURED AND MOUNTED.

```
kiouazan@kiouazan:/etc$ cd fshab
-bash: cd: fshab: No such file or directory
kiouazan@kiouazan:/etc$ cat fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devices
# that works even if disks are added and removed. See fstab(5).
#
# systemd generates mount units based on this file, see systemd.mount(5).
# Please run 'systemctl daemon-reload' after making changes here.
#
# <file system> <mount point> <type> <options> <dump> <pass>
# / was on /dev/sda1 during installation
UUID=67c03e1d-a487-4080-9d02-105e1df4a6fb / ext4 errors=remount-ro 0 1
# swap was on /dev/sda5 during installation
UUID=9547050f-a4b2-4263-85db-5598e4de9001 none swap sw 0 0
/dev/sr0 /media/cdrom0 udf,iso9660 user,noauto 0 0
```



3. PORT FORWARDING AND SSH ACCESS

Service réseau	Port de la VM	Port sur la station Linux	Exemple d'utilisation depuis la station Linux
SSH	22	2222	\$ ssh toto@localhost -p 2222
HTTP	80	8080	http://localhost:8080/
HTTPS	443	4443	https://localhost:4443/
PostgreSQL	5432	5432	\$ psql -h localhost -U postgres postgres

CHECK FOR THE ABSENCE OF THE XORG SERVER :

- Verify that the Xorg server is not installed on your machine using the following command:

```
DPKG -L | GREP XORG
```

- Why it's not installed:

This is a web server (Apache/PHP), so no graphical interface is needed. A minimal setup improves performance and security.



3.1 WEB SERVER DATABASE MANAGEMENT SETUP

Installation Guide: Apache, PostgreSQL, PHP, and PhpPgAdmin on Debian 12

APACHE 2

`apt install apache2`

`systemctl status apache2`

`systemctl start apache2`

VERIFY APACHE USING TELNET

```
$ TELNET LOCALHOST 80
TRYING ::1...
CONNECTED TO LOCALHOST.
ESCAPE CHARACTER IS '^]'.
HEAD / HTTP/1.0
```

You should get the following response:

```
HTTP/1.1 200 OK
```

```
root@klouazani/home/klouazani:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-03-24 13:49:48 CET; 3min 48s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1652 (apache2)
    Tasks: 55 (limit: 4642)
   Memory: 9.3M
      CPU(s): 3.5ms
   CGroup: /system.slice/apache2.service
           └─1652 /usr/sbin/apache2 -k start
             1656 /usr/sbin/apache2 -k start
             1657 /usr/sbin/apache2 -k start

Mar 24 13:49:48 klouazani systemd[1]: Starting apache2.service - The Apache HTTP Server
Mar 24 13:49:48 klouazani apache2[1648]: AH00558: apache2: Could not reliably determine
Mar 24 13:49:48 klouazani systemd[1]: Started apache2.service - The Apache HTTP Server
```

PHP

`apt install php libapache2-`

`mod-php`

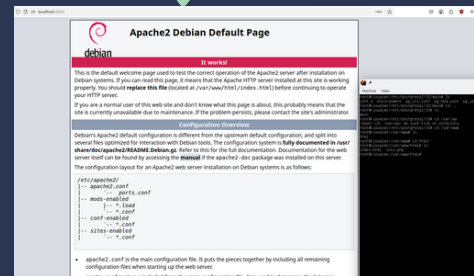
Then, open a browser on the host machine and navigate to:

You should see a page with details about your PHP installation.

VERIFY PHP INSTALLATION

CREATE A PHP INFO FILE IN /VAR/WWW/HTML/ WITH THE FOLLOWING CONTENT:

```
<?PHP
PHPINFO();
PHPINFO(INFO_MODULES);
?>
```



4. PHPPGADMIN

Install PhpPgAdmin

Consult PhpPgAdmin Documentation

Visit the Debian Wiki for PhpPgAdmin.

MODIFY PHPPGADMIN CONFIGURATION

IN THE FILE CONNECTION.PHP, CHANGE THE FOLLOWING LINE:

```
CASE '14': RETURN 'POSTGRES'; BREAK;

// CHANGE IT TO:

CASE '15': RETURN 'POSTGRES'; BREAK;
```

```
//$conf['servers'][1]['pg_dump_path'] = 'C:\\Program Files\\PostgreSQL\\8.0\\bin\\pg_dump.exe';
//$conf['servers'][1]['pg_dumpall_path'] = 'C:\\Program Files\\PostgreSQL\\8.0\\bin\\pg_dumpall.exe';

$conf['servers'][0]['desc'] = 'PostgreSQL';
$conf['servers'][0]['host'] = 'localhost';
$conf['servers'][0]['port'] = 5432;
$conf['servers'][0]['sslmode'] = 'allow';
$conf['servers'][0]['defaultdb'] = 'postgres';
$conf['servers'][0]['pg_dump_path'] = '/usr/bin/pg_dump';
$conf['servers'][0]['pg_dumpall_path'] = '/usr/bin/pg_dumpall';
```



3.2 POSTGRESQL

PostgreSQL Example Setup for more screen

```
kioouazan@kioouazan:/etc$ systemctl status postgresql
• postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Thu 2025-04-17 12:28:39 CEST; 7min ago
   Process: 605 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 605 (code=exited, status=0/SUCCESS)
   CPU: 1ms
```

01

Consult PostgreSQL
Documentation and
install

APT INSTALL POSTGRESQL

02

Verify PostgreSQL
Installation

SU - POSTGRES

03

Create a User and
Database in PostgreSQL

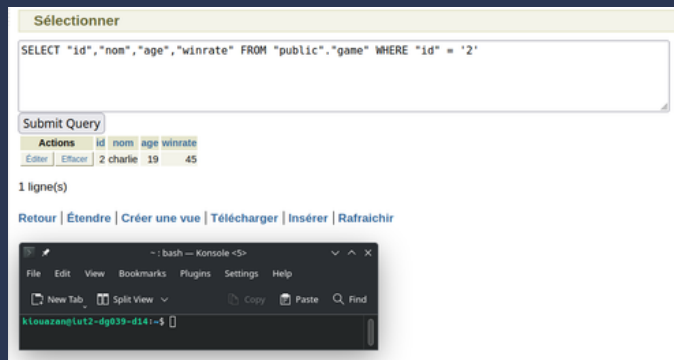
PSQL -L

04

Create a Table and
Insert Data

05

Configure PostgreSQL
for Remote Access



DATABASE CONFIGURATION AND USAGE

- CRÉE UN UTILISATEUR
CREATE USER KIOUAZAN WITH ENCRYPTED PASSWORD
'KIOUAZAN';
CREATE ROLE

- CRÉE UNE BASE DE DONNÉE POUR UN UTILISATEUR :
CREATE DATABASE MA_BASE WITH OWNER KIOUAZAN;
CREATE DATABASE

- CE CONNECTER A « MA_BASE » :
\C MA_BASE
YOU ARE NOW CONNECTED TO DATABASE "MA_BASE" AS
USER "POSTGRES".

```
kioouazan@postgres=> \l
```

Name	Owner	Encoding	Collate	List of databases Ctype	ICU Locale	Locale Provider	Access privileges
ma_base	kioouazan	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
postgres	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
template0	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres postgres=CTc/postgres +
template1	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres postgres=CTc/postgres +

(4 rows)

```
ma_base=# \set PROMPT1 '%n@%/%R '
```



4. LOGIN WITH SSH KEY

1. Generate an SSH Key Pair (on your host machine)

SSH-KEYGEN

Press Enter at each prompt to accept the defaults (RSA key, save to ~/.ssh/id_rsa, and no passphrase). This will create:

- Private key: ~/.SSH/ID_RSA
- Public key: ~/.SSH/ID_RSA.PUB

2. Copy the Public Key to Your Debian VM

On the server, open (or create) the authorized_keys file:

- Private key:
 - Located on your local machine in the ~/.ssh folder
- Public key:
 - Should be copied into the authorized_keys file on the remote machine (VM) in this path: '~/.ssh/authorized_keys'

NANO ~/.SSH/AUTHORIZED_KEYS
//COPY THE ENTIRE CONTENT OF THE PUBLIC KEY.

```
kiouazan@kiouazan:~$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kiouazan/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kiouazan/.ssh/id_rsa
Your public key has been saved in /home/kiouazan/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:Ig6Asv0Zf3lCL/PqYXVcDswzGebzx2gtjLTunff//UGE kiouazan@kiouazan
The key's randomart image is:
+---[RSA 3072]-----+
|      o      |
|.      = o   |
|+      @E.   |
|.o      +.%. |
|o . . . S  o B B|
| o = . . . o * +o|
| + . o .o.. +..|
| . o o.o.. ..|
| .. .o=.  o|
+---[SHA256]-----+
```

```
kiouazan@iut2-dg039-d14:~/.ssh$ ssh kiouazan@localhost -p 2222
Linux kiouazan 6.1.0-32-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.129-1 (2025-03-06) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Apr 17 16:20:39 2025 from 10.0.2.2
kiouazan@kiouazan:~$
```



5. POSTGRESQL EXAMPLE SETUP

USER ROLES OVERVIEW

username	usesysid	iscreatedb	isuser	isbypassrls	password	valuntil	useconfig
postgres	1	1	1	1	SCRAM-SHA-256\$4096:10uK1b0uP55g9mCTLm--\$5v6m1D158uyf0g3b0837850CF8:uohf2888Uu--cKPCrU7C9K1CC9211DyYrYrYcd:sdmEvdu/9d2ph-		
postgres	1	1	1	1	SCRAM-SHA-256\$4096:10uK1b0uP55g9mCTLm--\$5v6m1D158uyf0g3b0837850CF8:uohf2888Uu--cKPCrU7C9K1CC9211DyYrYrYcd:sdmEvdu/9d2ph-		
kiouazan	1	1	1	1	SCRAM-SHA-256\$4096:10uK1b0uP55g9mCTLm--\$5v6m1D158uyf0g3b0837850CF8:uohf2888Uu--cKPCrU7C9K1CC9211DyYrYrYcd:sdmEvdu/9d2ph-		

CREATE USER AND DATABASE

```
postgres=# CREATE USER kiouazan WITH ENCRYPTED PASSWORD 'kiouazan';
CREATE ROLE
postgres=# CREATE DATABASE ma_base WITH OWNER kiouazan;
CREATE DATABASE
```

```
kiouazan@iut2-dg027-d02:~$ S2.03-lance-machine-virtuelle
Image trouvée sur le serveur erebus4:
/users/Stockage-HDD/Images-kvm/S2.03/Images/Debian-S2.03-kiouazan.img
```

LIST OF DATABASES

```
postgres@kiouazan:~$ psql -l
```

Name	Owner	Encoding	Collate	Ctype	ICU Locale	Locale Provider	Access privileges
postgres	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
template0	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres + postgres=CTc/postgres

```
(3 rows)
```

DATABASE VERIFICATION

```
kiouazan@postgres=> \l
```

Name	Owner	Encoding	Collate	Ctype	ICU Locale	Locale Provider	Access privileges
ma_base	kiouazan	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
postgres	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	
template0	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres + postgres=CTc/postgres
template1	postgres	UTF8	en_US.UTF-8	en_US.UTF-8		libc	=c/postgres + postgres=CTc/postgres

```
(4 rows)
```

CONNECT AND LIST TABLES

```
kiouazan@iut2-dg039-d14:~$ psql -h localhost -U postgres postgres
```

```
Password for user postgres:
```

```
psql (15.12 (Debian 15.12-0+deb12u2))
```

```
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, compression: off)
```

```
Type "help" for help.
```

```
postgres@postgres=> \d
```

```
List of relations
```

Schema	Name	Type	Owner
public	sae203	table	postgres

```
(1 row)
```



6. LAST ADVICE.

Make sure to verify the storage location !

```
kiouazan@kiouazan:/etc$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           392M  496K   392M   1% /run
/dev/sda1       3.0G  1.6G   1.2G  57% /
tmpfs           2.0G  1.1M   2.0G   1% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
tmpfs           392M   0    392M   0% /run/user/1000
```

To maintain a secure LAMP stack (Linux, Apache, PostgreSQL, PHP) on Debian, it's essential to regularly monitor and patch potential vulnerabilities. Here's how to address software security issues:

- SUDO UFW ALLOW 22 80 443 SUDO UFW ENABLE
- APT UPDATE && SUDO APT UPGRADE
- DISABLE REMOTE ACCESS IF NOT NEEDED (LISTEN_ADDRESSES = 'LOCALHOST' IN POSTGRESQL.CONF).



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